

Eol Template Outline (+/- 1,500-2,500 words per Eol) – Due date CoB September 19

Please submit Eols to Masud Rana (MASUD.RANA@cgiar.org)

Section 0 – Title and contact Information

- Title of the proposed solution track

ACCELERATING THE COMMERCIALIZATION OF CLIMATE RESILIENT AND NUTRIENT DENSE BEAN VARIETIES in KENYA and MALAWI

- Title of the AoW2 Flagship that the solution track aligns with

2.1 Scaling Seed and Soil Fertility Management Systems for Resilient, Diverse, and Nutrition-Smart Farming

- Lead Scientist(s) / Contact Person(s) / Center(s)
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- Technical link to other Science Programs – name the corresponding Science Programs by name: Breeding for Tomorrow [Breeding for Tomorrow - CGIAR](#) and Sustainable Farming [Sustainable Farming - CGIAR](#)

Section 1 – Innovation Package, Value Proposition & Alignment

Purpose: To establish what the innovation is, who the customers and end-users are, why it matters, and how it connects to S4I's Flagships and Grand Challenges.

- **The problem to be solved:** (~150 words)

In Kenya and Malawi, common bean is a key staple and income source, yet seed value chains remain underdeveloped, limiting uptake of climate-resilient and nutrient-dense varieties. Despite the release of over 30 improved varieties, adoption remains low in both countries, with only 5–10% of seed being sourced from formal systems. Barriers include limited early generation seed (EGS) production, weak coordination between seed and grain markets, regulatory delays, and high seed costs. These challenges disproportionately affect women and youth, who face systemic barriers to accessing improved varieties and participating in seed markets. Strategic partnerships with CGIAR and Africa-wide organisations, such as AGRA, are essential to address EGS bottlenecks, regulatory harmonisation, and the design of inclusive seed systems. **AGRA has committed \$480,000 towards addressing last-mile constraints**, particularly the lack of demand-led planning, inclusive delivery models, and policy coherence, which are crucial to

unlocking scale and ensuring equitable impact for varieties with proven market potential and high IPSR scores.

Scaling ambition: (~150 words)

By 2028, this scaling track aims to reach **100,000 farmers—60%** of them women—across **50,000 hectares** in Kenya and Malawi, distributing 3,000 tons of certified seed through at least **five** licensed seed companies, which are already mobilised. Youth engagement will be prioritised through seed hubs, targeted training, and digitalisation of seed systems. These targets align with AoW2 Flagship 2.1 KPIs, contributing to increased adoption of CGIAR innovations, at **2000** youth employment opportunities and job creation in the seed and grain value chains, and improved nutrition. The scaling pathway builds on PABRA's bean corridor model and leverages the market systems expertise of partners, such as AGRA, ensuring bold yet achievable ambitions. Implementation will be anchored in existing infrastructure and committed partnerships to deliver inclusive, demand-driven impact at scale.

- **Innovation & Value Proposition: (~150 words)**

A bundling approach will be deployed scale climate-resilient, nutrient-dense bean varieties—such as Nyota, Faida, Waithera, KK-Rosecco 33, and Angaza in Kenya, and NUA45 and SUG131 in Malawi—alongside innovations in: (i) technology (improved varieties, agronomic practices), (ii) institutions (inclusive seed systems, market linkages), (iii) digital tools (seed traceability, farmer profiling), and (iv) finance (credit access, financial literacy). Delivery will follow PABRA's bean corridor model, integrating production and market expansion. The business logic is clear: seed companies benefit from licensing and market growth; farmers gain access to certified seed and market incentives; and digital and financial service providers expand their customer base. PABRA's MoUs with Pesira and Equity Bank enable delivery of financial literacy and digital services, especially to women and youth. The initiative complements CGIAR investments in seed systems, climate-smart agriculture, and nutrition-sensitive value chains, leveraging PABRA's inclusive, demand-led seed system and institutional readiness to deliver scalable, equitable impact.

Customers & Alignment: (~100 words)

This track brings together strategic partners: **KALRO** (Kenya) and **DARS** (Malawi) lead the varietal release process, EGS production, and seed system reforms. **AGRA** supports policy advocacy and market systems. The BRAINS project provides catalytic funding and gender-responsive tools. **TAAT** Bean Compact contributes to scaling and regional alignment. **Pesira** supports seed system digitalisation; **Equity Bank** delivers financial literacy and bundled services. **Farmers, SMEs, and offtakers** access improved seeds via inclusive MSPs, training, and co-financing. Engagement mechanisms such as licensing, profiling, and joint planning foster ownership. Aligned policies and donor strategies promote varietal turnover, private sector engagement, and climate-smart agriculture, advancing AoW2 Flagship 2.1 goals.

Evidence of Impact: (*~150 words*)

PABRA's demand-led seed systems model has demonstrated scalable impact in different bean-growing countries, including Kenya and Malawi. In Kenya, small seed packs and off-taker models have expanded access to improved varieties, particularly for women and young people, leading to increased productivity and income. In Malawi, MSPs have increased certified seed purchases and boosted willingness to pay by 25%. Transferability is evident, as demand-led seed systems models have been replicated elsewhere, e.g., in Zambia and Tanzania, and on other crops such as cowpea in Nigeria and groundnuts in Tanzania. However, challenges such as weak last-mile distribution and varietal awareness remain, which this track aims to address by ramping up EGS and certified seed generation through public-private partnerships and targeted promotion to accelerate farmer uptake in Malawi and Kenya.

Section 2 – Scaling Pathway, Sustainability, Partnerships & Risks

Purpose: To assess whether there is a credible pathway for the innovation to scale through public, private, or PPP delivery models — supported by sustainability plans, partner demand and commitments, financial leverage, and risk mitigation.

- **Scaling Pathway & Sustainability:**(*~250 words*)

The initiative will scale through a structured public-private partnership involving AGRA, ABC-PABRA, national research systems (KALRO and DARS), CGIAR centres (e.g., IITA, CIMMYT), and private seed and grain companies. AGRA and other donors will co-finance seed hubs and market development. ABC-PABRA and CGIAR centres will provide breeding expertise and technical backstopping. National systems will supply breeder and early generation seed (EGS), ensure quality assurance, and align efforts with national agricultural priorities. Private companies will invest in seed multiplication and distribution, incentivised by licensing revenues and growing market demand.

Producer groups, including cooperatives, will serve as last-mile distributors and grain suppliers, fostering circular value chains and local ownership. Implementation will be phased, with Year 1 (2026) focusing on seed availability and awareness through the establishment of a seed hub, multiplier training, and stakeholder alignment. Year 2 (2027) expands distribution, activates institutional markets (e.g., school feeding), and deploys demand forecasting tools. Year 3 (2028) consolidates gains by embedding seed scaling into national systems and policy frameworks.

The track champions a blended investment model combining public and private financing, supported by advisory services and digital tools for traceability, planning, and performance monitoring. Revenue streams include seed sales, licensing fees, and service contracts. Cost recovery is projected to reach 70% by 2028, enabling seed enterprises to sustain their operations beyond the project period. This approach ensures scalability, sustainability, and inclusive impact across the bean value chain.

- **Partnerships & Leverage:**(*~300 words*)

This initiative is built on a strong foundation of confirmed partnerships, each bringing tangible commitments and complementary capabilities to drive scale. AGRA has committed USD 480,000

through its Centre of Excellence for Seed Systems in Africa (CESSA)/AGRA to catalyse seed hub development, market activation, and policy alignment in Kenya and Malawi. ABC-PABRA, working in concert with CGIAR Centres such as IITA, CIMMYT, and *icipe* (non-CGIAR), will provide a multi-disciplinary innovation bundle—spanning breeding, crop resilience, seed system design, and regional scaling—ensuring the initiative draws from CGIAR’s broader research portfolio beyond beans. This collaboration also aligns with ongoing CGIAR initiatives, such as Accelerated Breeding and AoW2 Flagship 2.1, reinforcing coherence and avoiding duplication.

National research institutions—KALRO and DARS—are not only collaborators but also implementation anchors, actively engaged in EGS production, certification, and the integration of improved varieties into national seed strategies. The BRAINS project, co-implemented by ABC and *icipe* with support from Global Affairs Canada, contributes catalytic technical backstopping, with a strong emphasis on gender-responsive and climate-smart scaling. The AfDB TAAT High Iron Bean Compact, which is already reaching over 898,000 farmers and 169,000 hectares, provides a proven platform for scaling and policy leverage.

To strengthen last-mile delivery, PABRA has formalised partnerships with Pesira, a digital agriculture firm supporting farmer profiling and seed traceability, and Equity Bank, which delivers financial literacy and bundled services to smallholders—especially women and youth. On the commercial front, companies like **Spice World Ltd** are piloting aggregator and contract farming models to improve traceability and quality assurance in Kenya, signalling strong private sector interest in seed multiplication and distribution.

Together, these partnerships form a cohesive, well-resourced ecosystem that blends public research, private investment, and digital-financial innovation—positioning the initiative to deliver inclusive, scalable impact aligned with national priorities and CGIAR’s global mission.

- **Risks & Bottlenecks:** (~250 words)

Two priority risks could significantly affect delivery and sustained adoption of this initiative.

1. Fragmented seed system coordination

This poses a high-impact risk to varietal release, early generation seed production, and timely seed availability. Without synchronised planning, mismatches between seed supply and grain market demand can stall scaling. To mitigate this, the initiative will establish joint seed planning platforms in both countries, building on successful multi-stakeholder models from Malawi and Zambia, where seed producers, grain buyers, and public agencies co-develop seasonal plans. These platforms will improve coordination, accountability, and alignment with national seed strategies.

2. Weak demand forecasting and market anchoring

Inaccurate demand projections risk over- or under-production of seed, undermining private sector confidence and farmer trust. To address this, the initiative will deploy digital profiling tools (via Pesira) to capture real-time farmer demand and link this data to seed producers. Additionally, it will engage institutional buyers—such as school feeding programs and aggregators like Spice

World Ltd—to anchor demand and provide predictable offtake, improving planning accuracy and reducing market volatility.

Other operational risks—such as supply chain disruptions and staffing gaps—will be mitigated through decentralised seed production, local capacity building, and performance-based contracts. Gender and inclusion risks will be addressed through targeted training, gender-responsive packaging (e.g., small seed packs), and ensuring women and youth are represented in seed planning forums. These strategies are informed by PABRA's prior scaling experience, which shows that bundling seed with advisory services and engaging aggregators reduces exclusion and boosts adoption.

Section 3 – Responsible Scaling Trajectory & Achievable Scaling Target

Purpose: To define what can realistically be achieved with S4I support in the next 2–3 years — including achievable reach, milestones, resources, activities, and safeguards for responsible scaling.

- **Readiness & Responsible Scaling: (~150 words)**

The initiative demonstrates strong technical readiness, backed by PABRA's mid-term evaluation of the *Improving Bean Productivity and Markets in Africa (IBPMA)* project, which confirmed functional pluralistic seed systems and strong engagement of national agricultural research systems (NARS) in Kenya and Malawi. Commercial readiness is evident through signed MoUs and active offtaker interest, including Spice World Ltd piloting contract farming and traceability for Nyota and Angaza. Scaling will be guided by disaggregated KPIs for gender and youth, using participatory varietal selection, inclusive training, and support for women-led seed enterprises to address structural barriers. Environmental safeguards include stress-tolerant varieties, reduced synthetic input use, and training on soil fertility, intercropping, and post-harvest loss reduction, aligned with national climate-smart agriculture strategies. A Phase 1 readiness assessment will map institutional arrangements, risks, and entry points. A results-based MEL framework will include reflection sessions, user surveys, and digital tools (e.g., via Pesira) to enable adaptive management and continuous improvement.

Achievable target, Resources & Activities: (~250 words)

Year	Key Activities	Specific Indicators	Means of Verification	Frequency	Responsible Entity
2026	- Customise existing partnerships and sign six new MOUs - Establish/strengthen MSPs in Kenya and Malawi - Conduct readiness assessment - Identify seed hub sites and youth	- 4 partnerships customised - 6 MOUs signed - 2 MSPs established - Readiness assessment completed - Seed hub	- Signed MOUs - MSP minutes - Assessment report - Seed hub plans - EGS forecast	Quarterly	PABRA, AGRA, CGIAR centres

	groups - Initiate EGS planning and production	planning and EGS production initiated			
2027	- Train 100,000 farmers (40% women) - Deploy 2 financial tools to support beneficiaries - Launch 2 digital forecasting tools	- 100,000 farmers trained - 2 finance tools deployed - 2 digital tools operational	- Training records - Finance reports - Tool analytics	Biannual	Seed companies, NGOs, Equity Bank, Pesira
2028	- Facilitate seed sales via contracts - Operationalise 10 youth-run seed hubs - Integrate 5 varieties into national systems - Expand bundled services to 50,000 farmers Scaling the model/approach to other crops and countries	- 100,000 MT seed sold - 10 hubs operational - 5 varieties scaled - 50,000 farmers reached	- Contracts - Hub reports - Seed catalogues - Equity & Pesira reports	Quarterly	ABC-PABRA, KALRO, DARS, Equity Bank, Pesira

2026 budget: \$225,000

- **Timeline, Milestones & Monitoring:** (~200 words)

Monitoring will be anchored in a results-based framework using MELIA tools, with baseline assessments in 2026, midline in 2027, and endline in 2028 to track adoption, gender inclusion, and market outcomes. In 2026, seed hubs will be launched, and 500 service providers will be trained to establish a foundation for scaling. By 2027, the initiative is expected to expand to 50,000 hectares under improved bean management and reach 100,000 farmers, with disaggregated data capturing the participation of women and youth. In 2028, the focus will shift to consolidating gains and embedding scaling into national systems. The initiative is expected to create over 2,000 direct and indirect job opportunities across seed production, distribution, training, and digital services.

Digital dashboards will enable real-time tracking, reduce reporting costs, and support adaptive management. Participatory monitoring—via farmer feedback loops and aggregator reporting—will ensure credibility and responsiveness. Data collection and analysis will be led by PABRA's M&E specialist, supported by country focal agronomists and seed companies. Quarterly reviews

will refine strategy, address bottlenecks, and inform policy engagement, ensuring learning drives performance and inclusive scaling.

Optional Annexes

- IPSR Innovation Profile (if available)
- Scaling Readiness assessment results
- Scaling Scan results (if available)
- Prior project reports
- Media or other evidence of prior scaling work that can be accelerated
- MELIA studies, impact assessments, reports, publications
- Partner commitments (MOUs, LOIs, letters of support)

Annex 1: Resources & Tools to Support EOI Preparation

Teams preparing EOIs are encouraged to draw on the following core resources. These tools are directly aligned with the S4I operational criteria and will help ensure EOIs are evidence-based and robust:

Core Resources

1. **Scaling Scan**
 - Practical tool to assess strengths and weaknesses across 10 “scaling ingredients.”
 - Helps identify gaps in demand, finance, governance, partnerships, or capacities.
 - Resource: [Scaling Scan](#)
2. **Innovation Packages and Scaling Readiness (IPSR)**
 - CGIAR’s framework for assessing innovation maturity (readiness) and use, and for identifying bottlenecks and strategies for scaling.
 - Can provide a Scaling Readiness Level (SRL) and bottleneck map useful for Criterion 3.
 - Resource: [CGIAR IPSR Overview](#)
3. **Business Model Canvas (BMC-lite)**
 - Adapted version of the BMC for development contexts.
 - Useful for thinking through value proposition, partnerships, cost structures, and sustainability.
 - Resource: [Business Model Canvas](#): Templates and examples available via AoW2 co-leads.
4. **Responsible Scaling Guidance**
 - [GenderUp](#): A tool to integrate gender and inclusion into scaling design, helping identify trade-offs and opportunities for equitable, responsible scaling
 - Helps applicants strengthen Criterion 3 (Responsible Scaling safeguards).
5. **MELIA (Monitoring, Evaluation, Learning, Impact Assessment) Evidence**
 - [Past MELIA studies](#), adoption surveys, or bilateral project evaluations are valuable sources of impact evidence.
 - Resource: Contact MELIA focal points within your Center for relevant reports and datasets.

References

PABRA. (2022). Scaling improved bean varieties across Africa. Pan-Africa Bean Research Alliance. <https://www.pabra-africa.org>

Mukankusi, C., et al. (2019). Enhancing bean productivity and market access in Tanzania. *Journal of Agricultural Development*, 12(3), 45–58.

CGIAR Gender Platform. (2024). Gender-responsive climate-smart agriculture: Case studies from Kenya. <https://gender.cgiar.org>